

Gaze-Controlled Virtual Keyboard

Eye Tracking · Blink Detection · Predictive Text

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Problem

Individuals with ALS and Locked-In Syndrome lose speech and motor function – yet many retain eye movement and full cognitive ability.

EXISTING TECHNOLOGY ISSUES

► Expensive

\$800 – \$20,000 for proprietary hardware

► Slow

Sip-and-puff & switch scanning are tedious

► Inaccurate

Low-cost trackers suffer from noise & drift

► Inaccessible

Requires specialized equipment & setup

THE NEED

A low-cost, reliable, hands-free communication system using only consumer hardware – specifically a standard webcam.

Goal: Enable real-time typing through gaze and blink alone, without expensive eye-tracking peripherals.

Background

// Eye Tracking as Input

Gaze-based interaction is natural and intuitive for users who cannot operate traditional input devices.

// Hardware Barrier

Traditional systems rely on expensive infrared trackers or head-mounted equipment, limiting accessibility.

// Webcam Opportunity

Modern webcams + browser-based ML (WebGazer, MediaPipe) enable low-cost, software-only solutions.

// Multimodal Potential

Combining gaze, blink, and predictive text can compensate for individual weaknesses of each modality.

Existing Solutions

► Eye-Tracking Keyboards

Accurate but cost \$800-\$20K; require calibration

► Sip-and-Puff

Reliable but extremely slow; requires mouthpiece

► Switch Scanning

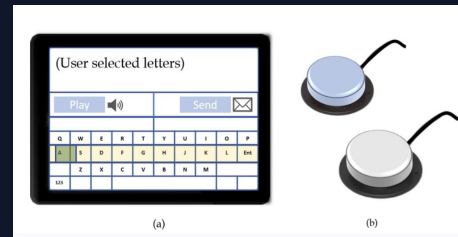
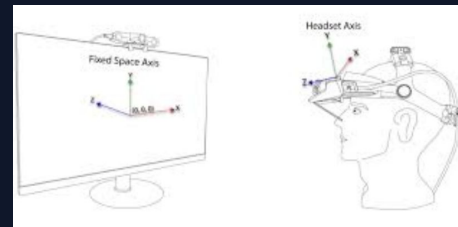
Single-button cycling; only viable with one remaining motion

► Head Tracking

Effective for limited neck mobility; causes fatigue over time

► Speech Recognition

Fast when it works; useless for non-verbal patients



Common tradeoff: cost ↓ → speed ↓ or accuracy ↓

Our Approach

A fully web-based gaze-controlled keyboard using only a standard webcam.

EYE TRACKING

WebGazer

Real-time gaze estimation from webcam

BLINK DETECTION

MediaPipe FaceMesh

Facial landmark detection for blink analysis

PREDICTIVE TEXT

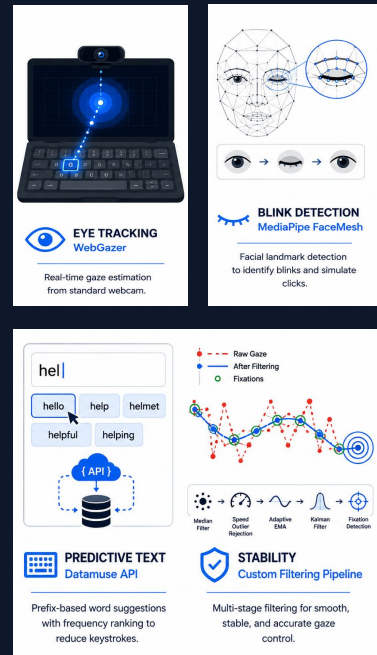
Datamuse API

Prefix-based word suggestions with frequency weighting

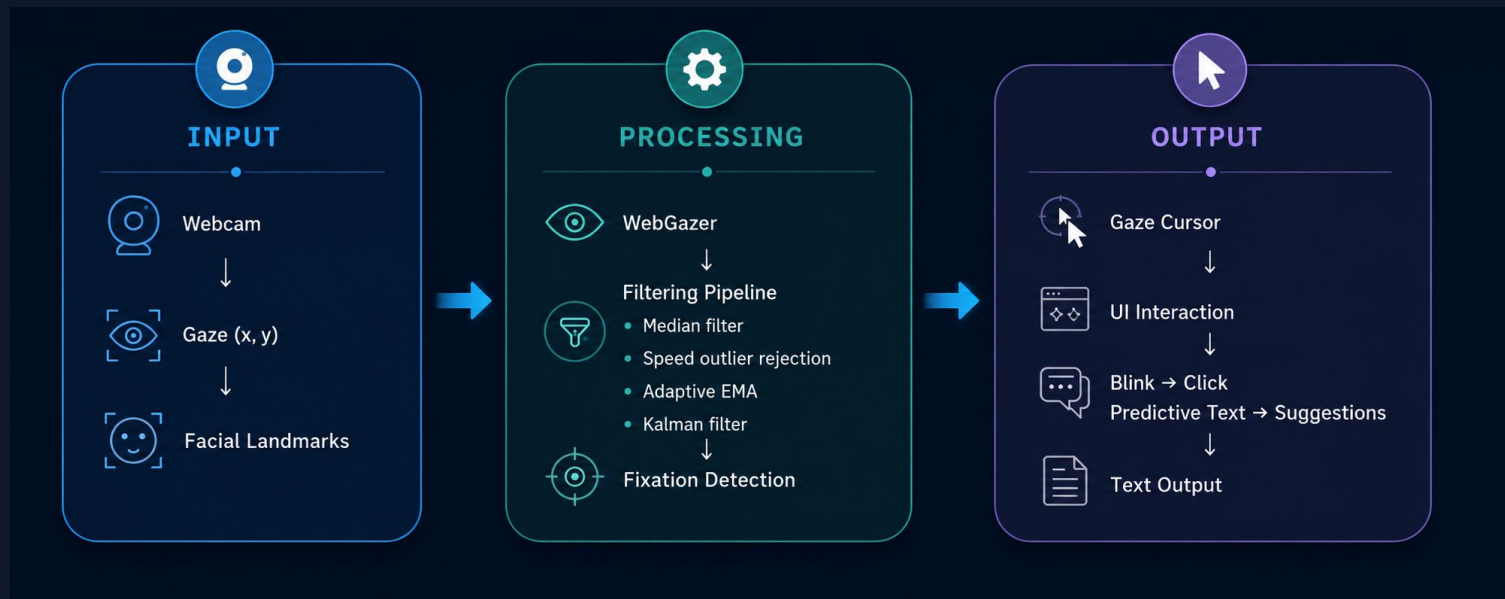
STABILITY

Custom Filtering Pipeline

Median + speed + adaptive EMA + Kalman filters



System Architecture



FaceMesh runs in parallel for blink detection → click simulation

Eye Tracking Improvements

Raw WebGazer output is noisy and unstable. A multi-stage filtering pipeline was implemented:

// Median Filter

5-point sliding window reduces high-frequency jitter

// Speed Outlier Rejection

Discards gaze jumps exceeding velocity threshold

// Adaptive EMA

Dynamic smoothing: fast = responsive, slow = stable

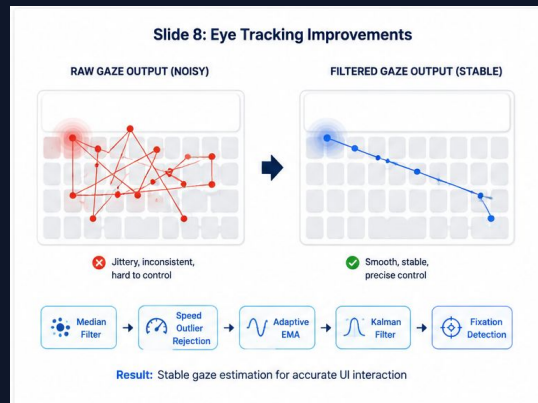
// Kalman Filter

Models gaze as dynamic system; optimally fuses prediction + measurement

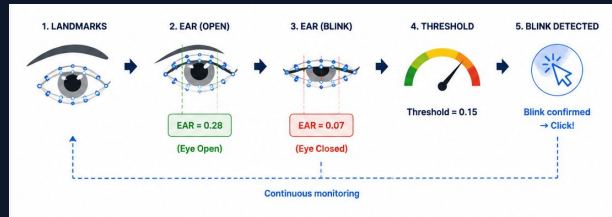
// Fixation Detection

Classifies fixations vs. saccades; prioritizes stable gaze for interaction

Result: Stable, controllable cursor movement suitable for UI interaction



Blink Detection



Fully hands-free click input using MediaPipe FaceMesh.

1. Landmark Extraction

FaceMesh provides precise eye region coordinates in real time.

2. EAR Calculation

Eye Aspect Ratio computed from specific landmark pairs around each eye.

3. Baseline Calibration

Open-eye EAR collected during initialization; threshold = fraction of baseline.

4. Blink Logic

Multiple consecutive frames below threshold + cooldown period → valid blink.

5. Click Simulation

Blink triggers mouse click at current filtered gaze position.

Predictive Text

Reduces keystrokes and improves typing speed via the Datamuse API.

→ Extract Prefix

Current incomplete word is parsed from the text input field.

→ API Query

Datamuse queried with prefix match+frequency metadata parameters.

→ Filter & Rank

Suggestions validated against prefix, ranked by frequency × relevance.

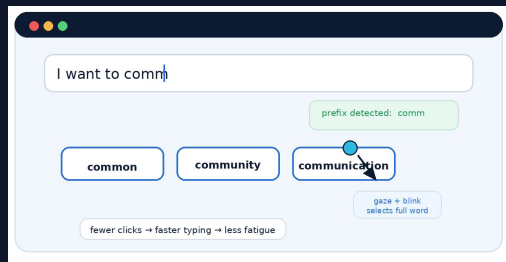
→ Display

Top suggestions shown in enlarged buttons above the keyboard.

→ Selection

User gazes + blinks on a suggestion → full word replaces the prefix.

Impact: Fewer selections per word → faster, less fatiguing typing



User Interface

Designed specifically for gaze-based interaction – every pixel matters.

- ▶ **Larger Keyboard Keys**

Bigger target areas reduce precision demands and selection errors.

- ▶ **Compact Text Box**

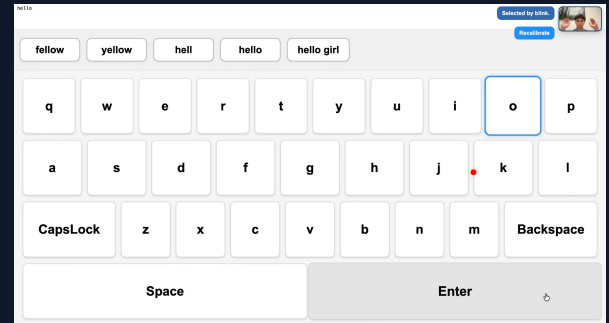
Frees screen real estate for the keyboard and suggestion bar.

- ▶ **Enlarged Suggestion Buttons**

Predictive text options are easy to target with gaze.

- ▶ **Minimal Visual Clutter**

Clean layout reduces cognitive load during prolonged use.



Small UI changes → large usability gains for gaze interaction

DEMO

Live typing with gaze and blink detection

Predictive text suggestions in action

Key Innovation

► Hardware Accessibility

Uses only a standard webcam – no proprietary eye-tracking hardware needed.

► Multimodal Fusion

Combines gaze tracking + blink detection + predictive text in one unified pipeline.

► Behavior-Aware Filtering

Fixation detection distinguishes intentional focus from random saccades.

► Adaptive Smoothing

EMA coefficient adjusts dynamically based on gaze speed – no lag vs. stability tradeoff.

► Practical Impact

Directly improves accessibility for ALS/LIS patients and others with motor impairments.

Challenges

⚠ Noisy Gaze Data

WebGazer output has inherent jitter; requires aggressive filtering.

⚠ Lighting Sensitivity

FaceMesh and WebGazer accuracy degrade in poor or changing light.

⚠ Calibration Drift

Gaze accuracy drifts over time; needs periodic recalibration.

⚠ Blink Detection Tuning

Threshold varies per user; false positives/negatives possible.

⚠ Smoothness vs. Responsiveness

Aggressive filtering reduces lag but can feel sluggish; adaptive EMA balances this.

Results

✓ Real-Time Gaze Tracking

Achieved sub-frame-latency gaze estimation using WebGazer + webcam.

✓ Stable Cursor Control

Layered filtering reduced jitter significantly; cursor usable for precise UI targeting.

✓ Reliable Blink Detection

FaceMesh EAR-based detection with user-specific calibration; low false-positive rate.

✓ Faster Typing

Predictive text cut keystrokes per word; suggestion bar integrated seamlessly.

✓ Functional System

End-to-end hands-free typing demonstrated with only consumer hardware.

Proof of concept: Low-cost gaze interaction is viable for assistive technology

Q & A

Questions?

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